

Do Organohalogen Contaminants Contribute to Histopathology in Liver from East Greenland Polar Bears (*Ursus maritimus*)?

.

In East Greenland polar bears (*Ursus maritimus*), anthropogenic organohalogen compounds (OHCs) (e.g., polychlorinated biphenyls, dichlorodiphenyltrichloroethane, and polybrominated diphenyl ethers) contributed to renal lesions and are believed to reduce bone mineral density. Because OHCs are also hepatotoxic, we investigated liver histology of 32 subadult, 24 adult female, and 23 adult male East Greenland polar bears sampled during 1999–2002. Light microscopic changes consisted of nuclear displacement from the normal central cytoplasmic location in parenchymal cells, mononuclear cell infiltrations (mainly portally and as lipid granulomas), mild bile duct proliferation accompanied by fibrosis, and fat accumulation in hepatocytes and pluripotent Ito cells. Lipid accumulation in Ito cells and bile duct hyperplasia accompanied by portal fibrosis were correlated to age, whereas no changes were associated with either sex or season (summer vs. winter). For adult females, hepatocytic intracellular fat increased significantly with concentrations of the sum of hexachlorocyclohexanes, as was the case for lipid granulomas and hexachlorobenzene in adult males. Based on these relationships and the nature of the chronic inflammation, we suggest that these findings were caused by aging and long-term exposure to OHCs. Therefore, these changes may be used as biomarkers for OHC exposure in wildlife and humans. To our knowledge, this is the first time liver histology has been evaluated in relation to OHC concentrations in a mammalian wildlife species, and the information is important to future polar bear conservation strategies and health assessments of humans relying on OHC-contaminated food resources.